
01 A VISION FOR WHAT COMES NEXT

Preparing students for the future of AI Product Development

A shared AI foundation for product teams and the future of technical education in Utah.

Alan Barth

Senior Program Director • MTECH

— 02 ABOUT ME

Nine years at MTECH. Still building.



STARTED HERE

Built the iOS program

- Started five new programs, beginning with iOS App Development.
- Built a practical AI course for people who are just getting started.



THEN

Office of Teaching and Learning



TODAY

Senior Program Director

- Guided faculty development in AI and its responsible, ethical use.
- I love using AI to build useful things and make everyday life easier.

— 03 BEFORE AI

Product teams before AI

A simplified view: work moves between specialties, supported by partner teams.

INSIDE THE PRODUCT TEAM

Shared responsibility was narrow

01



Product

Decide what and why

02



UX Design

Shape the experience

03



Development

Build the system

04



Testing

Verify the result

05



Go live

Release, support, and learn

PARTNER TEAMS OUTSIDE THE CORE



Marketing



Graphic Design



IT



Data

— 04 TECHNOLOGY EDUCATION TODAY

Aligned Programs

These programs are taught today and align with how technology companies organize their work.



Data Technology



Digital Design



Digital Marketing and Analytics



iOS App Development



Networking and Cybersecurity



Software Quality Assurance

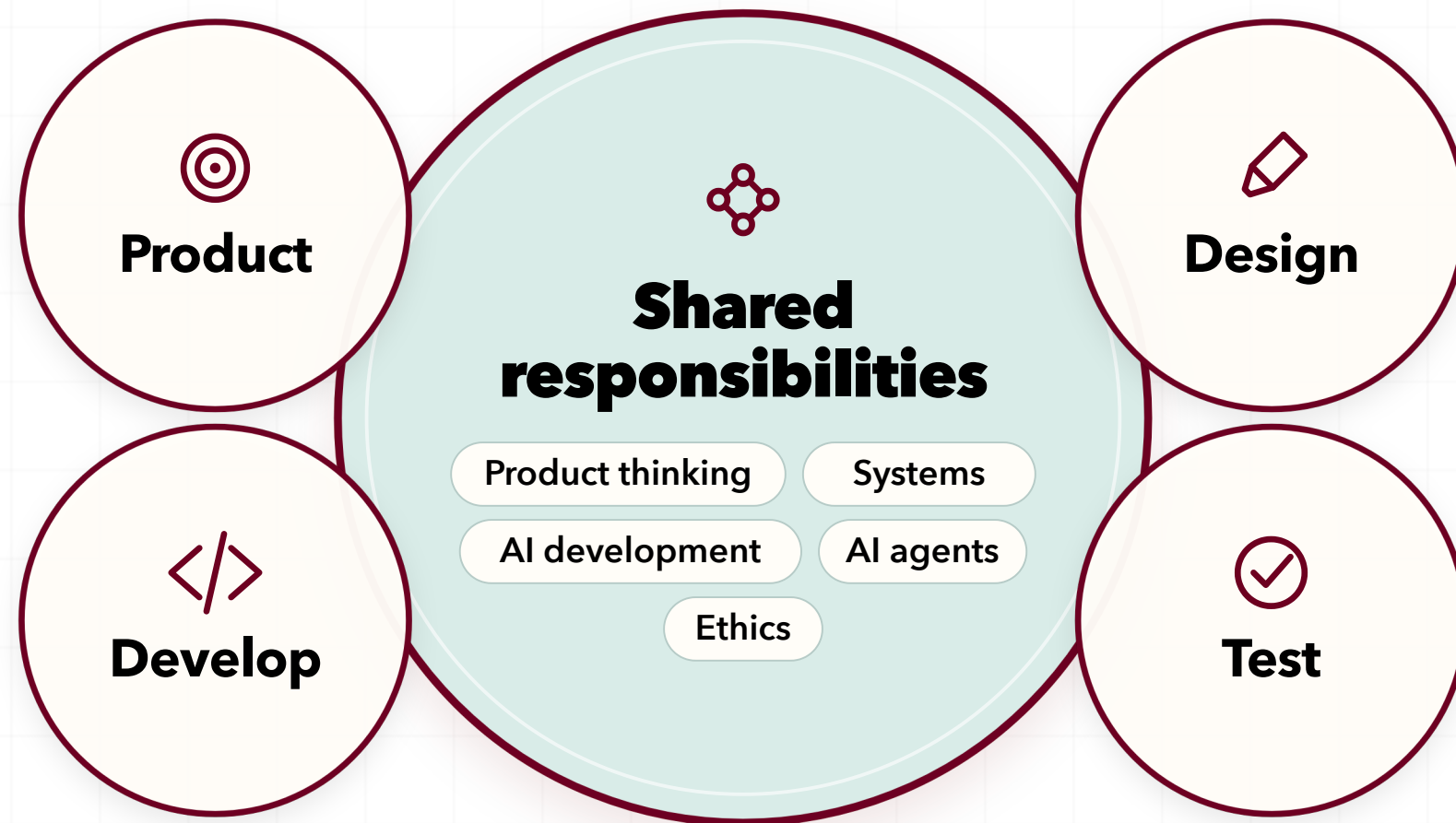


Web Programming and Development

— 05 THE AI-ENABLED TEAM

Product Teams with AI

With AI, more of the work is shared. The specialties remain distinct.



— 06 THE PROPOSED MODEL

One AI foundation. Three specialties.

Learn to plan, build, and verify products with AI. Then go deeper in one specialty.

EVERY STUDENT STARTS HERE • 360 HOURS

Shared Foundation Core

Product thinking

Systems

AI development

AI agents

Ethics



Design

Shape what gets built

360 hrs



Web

Build for the browser

360 hrs



iOS

Build for Apple platforms

360 hrs

360 hours of core + 360 in a specialty = 720 hours • 24 credits

— 07 THE SHARED CORE

Five courses. One AI foundation.

360 HOURS
12 CREDITS

Students learn to plan, build, and verify products with AI.

01

30h



Introduction to AI Product Development

STUDENT CREATES
First published site

02

60h



Product Thinking and Planning

STUDENT CREATES
Product brief

03

60h



Software Systems and Architecture

STUDENT CREATES
System plan

04

90h



Software Development with AI

STUDENT CREATES
Working product

05

120h



AI Agents and Automation

STUDENT CREATES
Agent workflow



Responsible AI in every course.

Protect privacy. Check AI output. Explain decisions.

— 08 THE SPECIALTIES

Shared AI skills. Specialist depth.

Students choose one 360-hour specialty: Design, Web, or iOS.

**12 CREDITS**

AI Product Development: Design

Shape the right product.

- ✓ Human-centered design
- ✓ Prototyping + design systems
- ✓ Research + product strategy

Product Designer • UX Designer • Junior
PM

**12 CREDITS**

AI Product Development: Web

Build complete web products.

- ✓ Modern front-end
- ✓ Back-end + APIs
- ✓ Cloud, security + deployment

Web Developer • Full Stack Developer

**12 CREDITS**

AI Product Development: iOS

Ship native mobile products.

- ✓ Swift + SwiftUI
- ✓ Data + cloud backends
- ✓ TestFlight + App Store

iOS Developer • Mobile App Developer

— 09 FROM A REAL PROJECT TO STUDENT LEARNING

A recipe directory. Shared AI skills.

I built a recipe directory app for the catering department.

How students could build it through the AI core



Understand
Help catering staff find recipes quickly.



Plan
Define recipe fields, categories, and search.



Build with AI
Create a searchable recipe directory.



Verify
Test search and check recipe details.



Automate
Suggest recipe tags for staff to review.

Then choose one specialty to go deeper



Design
Make recipes easy to find and follow.



Web
Build a browser-based recipe directory.



iOS
Create a native app for kitchen use.

Every student explains decisions, checks AI output, and improves the product.

— 10 ONE IDEA TO TAKE HOME

AI changed the work. Education can adapt.

 THE SHIFT

**AI expands the work every
role can do.**

Team boundaries now overlap.



 THE OPPORTUNITY

**Build a broad shared AI
core.**

Then go deep in Design, Web, or
iOS.



 THE RESULT

**Specialists who can build
across boundaries.**

Ready for AI-enabled product teams.



Download presentation slides

All 11 slides · PDF



Download course descriptions and objectives

Foundation core and all specialties · PDF

— 11 THE REQUEST

A next step for AI education in Utah.

Support moving forward with the AI Product Development program.



First cohort

August 2027

Proposed start



Time to complete

9 months

From start to finish



Program length

720 hours

Shared core + one specialty

MTECH would host the first cohort and share results with technical colleges across Utah.